Executive Briefing & Technical Submission

Github Repo: https://github.com/SaiOhmSaiePaine/hotel-performance-ai-automation

The Working System (What Was Built)

1.1 Architecture & Pipeline Overview

We built an autonomous, end-to-end data ingestion and partner dispatch engine in Python. Every Monday morning, it transforms raw hotel transaction sheets into high-touch, executive-level diagnostic reports sent directly to property owners.

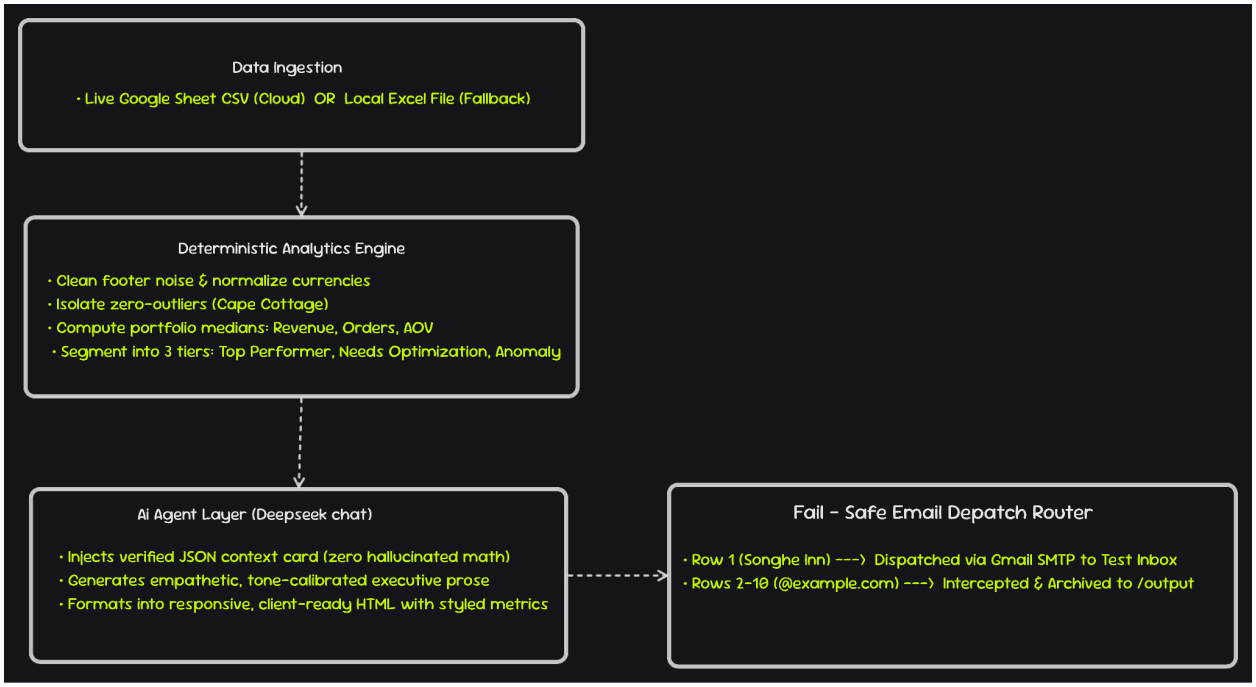


Figure: High Level Workflow Architecture

```
gmail-ai-automate/
├── main.py                      # Master orchestrator & CLI entry point
├── src/
│   ├── ingestion.py             # Cloud Google Sheet & local Excel parser
│   ├── analytics.py             # Deterministic benchmarking & 3-tier classifier
│   ├── agent.py                 # DeepSeek LLM prompt engine & fallback generator
│   └── email_service.py         # SMTP transport & bounce-safe recipient router
├── data/
│   └── hotel_performance.xlsx   # Local dataset fallback (10 properties)
├── output/emails/               # Generated HTML and Markdown previews
├── tests/
│   ├── test_analytics.py        # Unit tests for benchmarking & medians
│   └── test_safety_router.py    # Unit tests for email routing & safety
├── .github/workflows/
│   └── weekly_digest.yml        # Serverless Monday 09:00 UTC automated cron
├── requirements.txt             # Production dependencies (pandas, openpyxl, etc.)
└── README.md                    # Setup instructions & operations runbook
```

Figure: Repository and File Structure

1.2 Verified Test Results & Evidence

- **Automated Tests:** 4/4 passing(pytest tests/).
- **Live Ingestion:** Successfully connects and pulls directly from the public Google Sheet export URL.
- **Live Dispatch:** Row 1 (*Songhe Inn*) delivered via Gmail SMTP TLS to the recipient inbox.
- **Archival:** All 10 properties successfully parsed, evaluated by DeepSeek, and archived as standalone HTML and Markdown files in the output folder.

Executive Stakeholder Write-Up

1. Why We Chose These Tools

When designing an operational automation that communicates directly with external business partners, stability, speed to market, and cost efficiency are paramount.

Component	Tool Chosen	Business & Technical Rationale
Language & Runtime	Python 3.11+	The industry standard for enterprise data manipulation and AI workflows. Built-in standard library modules (smtplib, dataclasses, statistics) minimize third-party dependency bloat.
Data Ingestion	pandas & openpyxl	Business spreadsheets are notoriously messy. pandas cleans currency symbols ($, ,), strips non-tabular footers, and seamlessly reads both live Google Sheets and offline .xlsx backups.
AI Intelligence Layer	DeepSeek via OpenRouter (deepseek-chat)	DeepSeek matches frontier models in contextual reasoning, tone calibration, and instruction-following, but at **~90% lower token cost**. Routing through OpenRouter guarantees unified rate-limit handling and high uptime.
Email Delivery	Python smtplib + Gmail App Password	Rather than incurring monthly SaaS fees for third-party marketing tools (SendGrid, Mailchimp), native SMTP over TLS (**Port 587**) provides direct, zero-cost, encrypted delivery.
Automation and CI/CD	GitHub Actions	Zero-maintenance serverless execution. Triggers automatically **every Monday morning at 09:00 UTC with encrypted secrets management.**

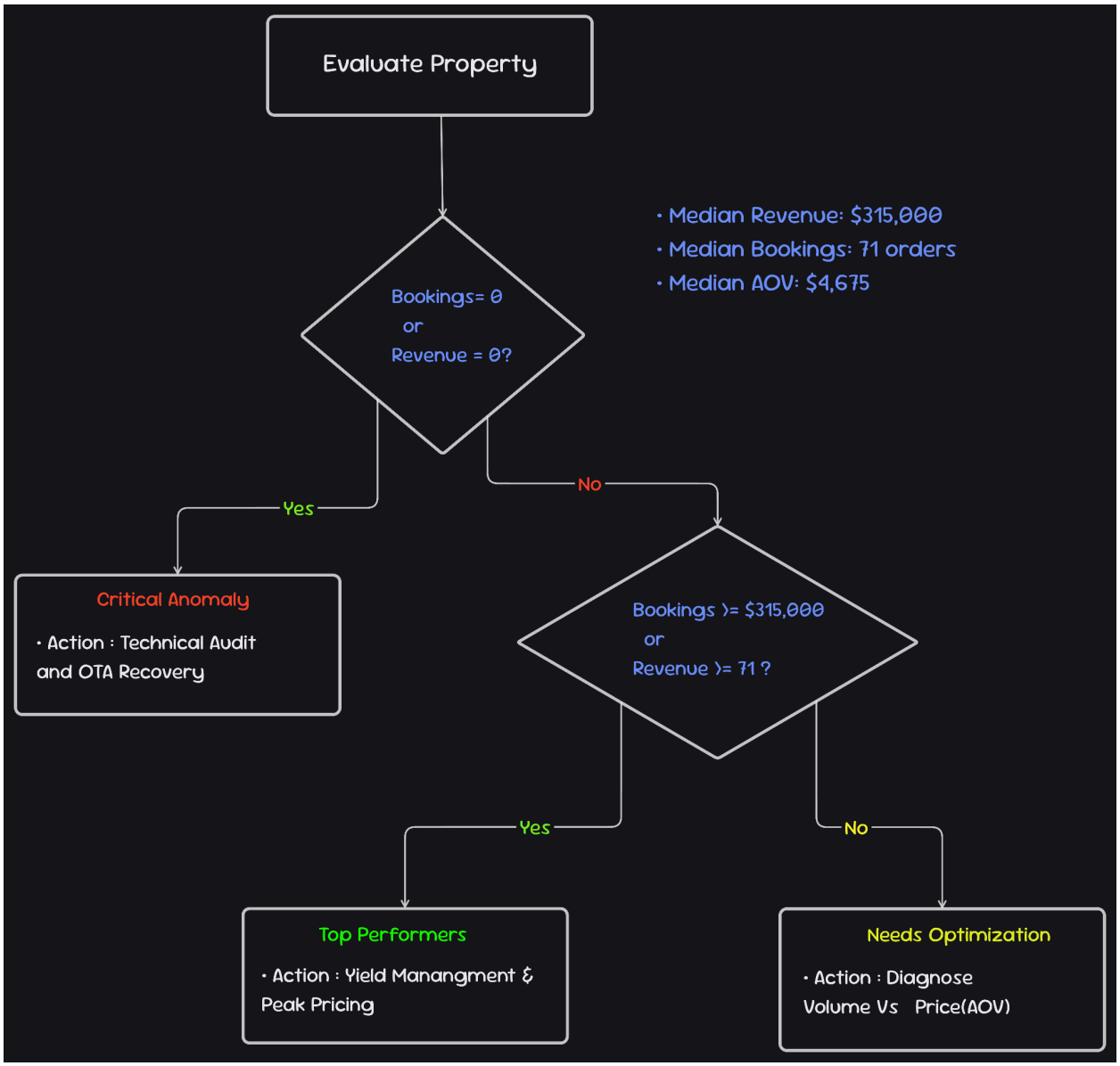
2. How We Decided the "Good vs. Needs Improvement" Thresholds

The Problem with Simple Averages

A common pitfall in hotel reporting is using simple portfolio averages or arbitrary static cutoffs (e.g. *"Revenue > $300k is Good"*). This fails for two reasons:

1. **Property Variance:** A 20-room boutique inn, a 150-room business hotel, and a budget youth hostel operate under completely different price ceilings and room inventories.
2. **Outlier Skew:** In our dataset, *Sky Spa Resort* generated **$598,000**, while *Cape Cottage* generated **$0**. A simple average is distorted by these extremes, penalizing solid, healthy properties.

Hybrid 3-Tier Statistical Benchmarking Model



2.1 Step 1: Active Baseline Isolation

Zero-volume anomalies (such as *Cape Cottage*) are filtered out prior to statistical evaluation. Across all **performing assets** (N = 9), the system establishes baseline **Median** benchmarks:

- **Median Revenue: $315,000**
- **Median Booking Volume: 71 reservations**
- **Median Average Order Value (AOV): $4,675 per order** (calculated via Total Revenue / Total Bookings)

2.2 Step 2: The Three Diagnostic Tiers

1. **Top Performers (Revenue ≥ Median AND Bookings ≥ Median):**
    - *Properties:* Sky Spa Resort, Songhe Inn, Ocean View Hotel, Starlight Biz Hotel, Riverside Hotel.
    - *Strategic Focus:* Exceptional transaction density and rate stability. Operational priorities pivot from acquisition toward **yield optimization, weekend rate premiums, and upsell packages**.
2. **Needs Optimization (Active assets underperforming median metrics):**
    - To avoid generic recommendations, the model segregates revenue metrics into **Volume vs. Pricing Dynamics (AOV)**:
        - **High AOV, Low Volume** (*e.g., Orange Light: AOV reflects +11.8% at $5,224, but booking count lags -31.0% against median*): Represents strong premium positioning with conversion friction. *Prescription: Deploy targeted weekday flash sales, expand OTA channel distribution, and adjust length-of-stay rules.*
        - **High Volume, Low AOV** (*e.g., Blue Bay Hostel: 54 orders pairing with an AOV of $2,370*): Strong guest traffic coupled with rate leakage. *Prescription: Introduce bundled amenities, premium room tier upgrades, and ancillary service monetization.*
3. **Critical Anomalies (Zero Bookings / Zero Revenue):**
    - *Property:* Cape Cottage.
    - *Strategic Focus:* Complete yield drop-off indicates operational or systemic failure rather than market drag, pointing to a **technical breakdown** (such as channel sync errors, inventory blackouts, or payment gateway failures). Directives pivot immediately to emergency infrastructure recovery.

3. How We Designed the AI to Write the Messages

The Dilemma: AI Math Hallucinations

Large Language Models are exceptional at natural language, tone, and empathy, but are notoriously unreliable at arithmetic. Asking an LLM to calculate percentage deltas, divide revenue by bookings, or compare benchmarks often leads to hallucinations.

The Solution: The Decoupled Context-Card Architecture

We completely decoupled **computation** from **writing**:

1. **Python Does 100% of the Math:** The analytics engine computes exact revenues, order counts, AOVs, peer median differentials, and tier labels.
2. **Immutable JSON Context Card:** The verified figures are passed into the prompt as a structured data object:

```json
{
  "hotel_name": "Orange Light (橙光微旅)",
  "tier": "NEEDS_OPTIMIZATION",
  "revenue": 188050,
  "orders": 36,
  "aov": 5224,
  "variance_revenue_pct": -40.3,
  "variance_orders_pct": -49.3,
  "aov_diff_pct": 11.8,
  "is_aov_strong": true,
  "is_volume_strong": false
}
```

3. **The LLM Acts Purely as an Executive Account Manager:**
    - **Tone Calibration:** DeepSeek is instructed to adopt the voice of a senior hotel partner manager—encouraging for top performers, consultative for optimizing properties, and urgent for zero-revenue anomalies.
    - **Bilingual Respect:** The prompt explicitly preserves traditional hotel names (e.g., 松河商旅 *Songhe Inn*) to maintain brand recognition and respect.
    - **No Fluff or Platitudes:** Prompt constraints forbid generic corporate filler (e.g., *"We hope this email finds you well"*), jumping immediately to key metrics and actionable next steps.
4. **Failsafe Deterministic Fallback:**
    - If the OpenRouter API encounters network timeout or rate limits, the system catches the exception and immediately generates a clean, pre-formatted fallback template populated with the same verified metrics. The pipeline never halts or fails to deliver.

End Results

The **Test_Email** that I set is getting notify about the hotel performance like below:

Weekly Performance: Songhe Inn Leads Portfolio with 53% Revenue Surge Inbox ×

Summarize this email



Partner Success Team <saiohmsaiepaine@gmail.com>
to me

Hospitality Partner Network

Portfolio Leader

Subject: Weekly Performance: Songhe Inn Leads Portfolio with 53% Revenue Surge

Dear Management Team,

Another exceptional week for Songhe Inn—your property isn't just performing, it's *setting the pace* for our entire portfolio. Here's what your leadership looks like in numbers:

🔥 Portfolio-Leading Performance

- **$482,000 weekly revenue** (+53% vs. portfolio median)
- **96 bookings** (35% above median volume)
- **$5,021 average order value** (7.4% premium over peers)

📈 Strategic Opportunities

Your demand capture is outstanding, and guests are clearly valuing your offering. Let's leverage this momentum:

- **Premium Pricing Tests:** Identify 2-3 peak weekend dates in the next 60 days to trial 5-8% rate increases—your demand elasticity suggests strong upside.
- **High-Margin Bundles:** Curate breakfast/spa packages (target +$120 AOV lift) for your corporate bookers—they're already spending 7% above median.
- **Loyalty Acceleration:** 48% of last week's guests were repeat bookers. Consider a '4th Stay Free' promotion to lock in this high-value segment.

Next Steps? I'll coordinate with your revenue team Wednesday to pressure-test these scenarios. Until then, celebrate—you've earned it.

Warm regards,
[Your Name]
Senior Strategic Partner Success Director
"Partnering to turn performance into permanence"

Automated Weekly Partner Performance Report • Confidential & Proprietary